Gaussian $p(\chi_{\rm eff}, \chi_p \mid m_1, q)$ $(k = 100)$

all q | $q \in [0.8, 1.0]$ | $q \in [0.3, 0.65]$

all m_1 | $m_1 \in [8, 12]\,M_\odot$ | $m_1 \in [16, 22]\,M_\odot$ | $m_1 \in [30, 40]\,M_\odot$ | $m_1 \in [55, 65]\,M_\odot$

χ_p | $\chi_{\rm eff}$

$\chi_{\rm eff} = 0$
$\chi_{\rm eff} = 0.2$
$\chi_p = 0.2$
$\chi_p = 0.5$

$P(q \in Q)$
$0.68^{+0.14}_{-0.41}$

$P(q \in Q)$
$0.14^{+0.11}_{-0.059}$

$P(m_1 \in M)$
$0.66^{+0.089}_{-0.1}$

$P(q \in Q \mid m_1 \in M)$
$0.82^{+0.15}_{-0.58}$

$P(q \in Q \mid m_1 \in M)$
$0.03^{+0.13}_{-0.028}$

$P(m_1 \in M)$
$0.069^{+0.044}_{-0.031}$

$P(q \in Q \mid m_1 \in M)$
$0.081^{+0.23}_{-0.071}$

$P(q \in Q \mid m_1 \in M)$
$0.65^{+0.21}_{-0.27}$

$P(m_1 \in M)$
$0.047^{+0.019}_{-0.015}$

$P(q \in Q \mid m_1 \in M)$
$0.75^{+0.14}_{-0.2}$

$P(q \in Q \mid m_1 \in M)$
$0.067^{+0.096}_{-0.051}$

$P(m_1 \in M)$
$0.0043^{+0.0043}_{-0.0023}$

$P(q \in Q \mid m_1 \in M)$
$0.04^{+0.16}_{-0.037}$

$P(q \in Q \mid m_1 \in M)$
$0.68^{+0.24}_{-0.35}$

$p(\chi_{\rm eff}, \chi_p \mid m_1, q)$